

Reduced-Rank Regression & Repeated Measures Analysis

Readings: Izenman 2008, Chapter 6; DSA 8020 Lectures 10 & 11 [\[Link\]](#); DSA 8070 Week 4 MANOVA

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Agenda

Recap of Multivariate Regression

Reduced-Rank Regression

Repeated Measures Analysis

Review of Multiple Linear Regression

Setup (univariate response, multiple predictors):

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n,$$

$$\varepsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$$

Classical workflow: model selection, least squares estimation, inference on β 's, diagnostics

Limitation: When there are *multiple* response variables that may be correlated, separate univariate models can be inefficient and miss joint structure

What is Multivariate Regression?

Goal: Predict a vector response $\mathbf{Y}$ (dimension q) from predictors $\mathbf{X}$ (dimension p).

Model:

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E}, \quad \mathbf{Y} \in \mathbb{R}^{n \times q}, \quad \mathbf{X} \in \mathbb{R}^{n \times p}, \quad \mathbf{B} \in \mathbb{R}^{p \times q}, \quad \mathbf{E} \in \mathbb{R}^{n \times q},$$

$$\mathbf{e}_i \stackrel{i.i.d}{\sim} N_q(\mathbf{0}, \Sigma)$$

Note:

- ▶ Running OLS separately on each column of $\mathbf{Y}$ = doing multiple regressions for each response
- ▶ This *ignores* correlation among responses
- ▶ $\Rightarrow$ Estimates of coefficients stay the same, but inference changes because of correlated errors

Why Use Multivariate Regression?

Gains from modeling responses jointly:

- ▶ Account for dependence among responses in **joint inference**
- ▶ Test whether predictors affect the **response vector as a whole**
- ▶ Exploit shared structure across responses through **dimension reduction**
- ▶ Build parsimonious models when responses are strongly related

Applications: repeated measurements, growth curves, multivariate outcomes, and other correlated-response problems

What Does “Reduced Rank” Mean Here?

Suppose q responses have regression coefficients collected in $\mathbf{B} = [\beta_1, \dots, \beta_q]$.

Full-Rank Model

Each response may depend on the predictors in a different way.

Reduced-Rank Model

Responses share a smaller number of underlying predictor effects.

$\text{rank}(\mathbf{B})$ can be as large as $\min(p, q)$.

$\text{rank}(\mathbf{B}) = r < \min(p, q)$.

Many distinct predictor–response relationships may be needed.

Only r directions describe the systematic predictor–response structure.

Reduced rank (low rank) = shared structure across responses, not fewer responses.

Reduced-Rank Regression (RRR) – Intuition

Suppose the coefficient matrix has reduced rank:

$\text{rank}(\mathbf{B}) = r < \min(p, q)$, p predictors, q responses

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E} = (\mathbf{XA})\mathbf{C} + \mathbf{E}, \quad \mathbf{B} = \mathbf{AC}, \quad \mathbf{XA}: r \text{ latent scores.}$$

$$\underbrace{\mathbf{X}}_{n \times p} \longrightarrow \underbrace{\mathbf{XA}}_{n \times r} \longrightarrow \underbrace{\widehat{\mathbf{Y}}}_{n \times q}$$

Instead of allowing p predictors to influence q responses arbitrarily, RRR forces those relationships through only r latent dimensions.

Mathematical Formulation of RRR

Model: $\mathbf{Y} = \mathbf{XB} + \mathbf{E}$, with $\text{rank}(\mathbf{B}) \leq r$

Rank-constrained least squares:

$$\min_{\mathbf{B}: \text{rank}(\mathbf{B}) \leq r} \|\mathbf{Y} - \mathbf{XB}\|_F^2$$

Key idea:

Start with ordinary multivariate regression, then project the fitted values to a lower-rank approximation via Singular Value Decomposition (SVD)

Why Does SVD Enter Reduced-Rank Regression?

Start with the unrestricted multivariate regression:

$$\widehat{\mathbf{Y}}_{\text{OLS}} = \mathbf{X} \widehat{\mathbf{B}}_{\text{OLS}}.$$

The fitted responses may contain more dimensions than we really need.

Full fitted response $\xRightarrow{\text{SVD}}$ Best rank- r approximation

SVD identifies the dominant directions of variation in $\widehat{\mathbf{Y}}_{\text{OLS}}$ and allows us to retain only the r most important directions.

RRR = fit first, then retain the dominant response directions.

Where Does RRR Fit among Dimension-Reduction Methods?

- ▶ **PCA**: finds low-dimensional structure within a set of variables
- ▶ **CCA**: finds directions that relate two sets of variables
- ▶ **RRR**: finds a low-dimensional structure specifically for predicting multiple responses from predictors

For now, focus on RRR as **multivariate regression with a low-rank constraint**. We will revisit the connections after PCA and CCA.

Estimation via Singular Value Decomposition (SVD)

Steps:

1. Fit full multivariate least squares:

$$\hat{\mathbf{B}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

2. Compute fitted values: $\hat{\mathbf{Y}} = \mathbf{X} \hat{\mathbf{B}}_{\text{OLS}}$

3. SVD of fitted values: $\hat{\mathbf{Y}} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$

4. Best rank- r approximation (retain top r singular values/vectors):

$$\hat{\mathbf{Y}}^{(r)} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top$$

5. Corresponding RRR estimator:

$$\hat{\mathbf{B}}_{\text{RRR}} = \hat{\mathbf{B}}_{\text{OLS}} \mathbf{V}_r \mathbf{V}_r^\top,$$

i.e., dimension reduction in the response space via $\mathbf{V}_r$

Choosing the Rank r

How many latent dimensions?

- ▶ Cross-validation: choose r with best predictive performance
- ▶ Information criteria: AIC, BIC (with appropriate df adjustments)
- ▶ Likelihood ratio tests (LRTs) under normality assumptions

Trade-offs:

- ▶ Underfit (r too small): bias
- ▶ Overfit (r too large): variance inflation

Simulation Example

Scenario: $p = 6$ predictors; $q = 4$ responses; true rank $r^* = 2$

Compare: separate OLS vs. RRR with $r = 2$ (correct), $r = 0, 1$ (underfit), $r = 3, 4$ (overfit).

- ▶ When responses are correlated, RRR typically outperforms separate OLS in prediction.
- ▶ Evaluate predictive error or estimation error as a function of rank.

Check an example in the R session

Practical Considerations & Assumptions

- ▶ Linearity is assumed; normality needed for classical tests, or rely on large-sample results
- ▶ Error covariance and equal-variance (homoscedasticity) assumptions matter for inference
- ▶ Multicollinearity in $\mathbf{X}$ or high correlation in $\mathbf{Y}$ can make estimation tricky
- ▶ Computation can be challenging when p, q are large compared to n
- ▶ Regularized approaches (ridge + rank, nuclear norm) help in high-dimensional settings

Extensions: Regularized Reduced-Rank Regression

We learned some [regularization techniques](#) in **8020 Week 6**

- ▶ **Reduced-Rank Ridge Regression:** combines dimension reduction (low-rank) with shrinkage
- ▶ **Nuclear-norm penalty:** a convex tool that encourages low-rank structure
- ▶ **Why useful?** More stable, handles noise better, and generalizes well when p, q are large but n is only moderate

Summary & Take-Home Points

- ▶ Multivariate regression extends regression to **multiple responses**
- ▶ Reduced-Rank Regression (RRR) exploits **shared structure** across responses for dimension reduction and potentially improved estimation
- ▶ RRR constrains the coefficient matrix:

$$\text{rank}(\mathbf{B}) = r < \min(p, q)$$

- ▶ Rank selection balances complexity: **too small** $\rightarrow$ underfitting; **too large** $\rightarrow$ overfitting
- ▶ **Extensions:** regularization for high-dimensional settings and reduced-rank methods for repeated-measures and other structured multivariate data

Why Are Repeated Measures Multivariate?

Suppose subject i is measured at four time points:

$$\mathbf{Y}_i = (Y_{i1}, Y_{i2}, Y_{i3}, Y_{i4})^T.$$

Each subject contributes **one multivariate response vector**, with correlated measurements over time.

- ▶ Repeated-measures methods differ mainly in how they model **within-subject dependence**.
- ▶ A key question is: **How much structure should we impose on the covariance?**

The goal is to balance **flexibility** and **parsimony** in modeling within-subject dependence.

Dog Experiment [Source: PSU STAT 505]

A completely randomized block design was carried out to determine the effects of 4 surgical treatments on coronary potassium in a group of 36 dogs. There are 9, 8, 9, and 10 dogs in each treatment group, respectively. Each dog was **measured at four different time points** (1, 5, 9, and 13 minutes) following one of four experimental treatments:

- ▶ Control - no surgical treatment is applied
- ▶ Extrinsic cardiac denervation immediately prior to treatment
- ▶ Bilateral thoracic sympathectomy and stellectomy 3 weeks prior to treatment
- ▶ Extrinsic cardiac denervation 3 weeks prior to treatment

We are looking at the treatment effect on the coronary sinus potassium levels

Notation of Approaches

Let Y_{ijk} be the potassium level for treatment i in dog j at time k :

- ▶ there are $a = 4$ treatments (i.e., $i = 1, 2, 3, 4$)
- ▶ n_i dogs received treatment i (therefore, there are $n_1 + \dots + n_a = 9 + 8 + 9 + 10 = 36$ dogs in total)
- ▶ $t = 4$, the number of observations over time (i.e., $k = 1, 2, 3, 4$)

Approaches

- ▶ Split-plot ANOVA
- ▶ MANOVA
- ▶ Mixed Models

Three Ways to Handle Within-Subject Dependence

Approach	Dependence Assumption	Main Trade-off
Split-plot ANOVA	Compound symmetry	Simple but restrictive
MANOVA	Unstructured covariance	Flexible but parameter-heavy
Mixed model	Specified structure (e.g., AR(1))	Flexible and parsimonious

The methods differ primarily in **how they model within-subject dependence**.

Approach 1: Split-plot ANOVA

Model: $Y_{ijk} = \mu + \alpha_i + \delta_{j(i)} + \beta_k + (\alpha\beta)_{ik} + \varepsilon_{ijk}$,

where

- ▶ α_i : effect of treatment i
- ▶ $\delta_{j(i)}$: random effect of dog j receiving treatment i
- ▶ β_k : effect of time k
- ▶ $(\alpha\beta)_{ik}$: treatment by time interaction
- ▶ ε_{ijk} : random error

Assumptions:

- ▶ $\varepsilon_{ijk} \stackrel{i.i.d.}{\sim} N(0, \sigma_\varepsilon^2)$
- ▶ $\delta_{j(i)} \stackrel{i.i.d.}{\sim} N(0, \sigma_\delta^2)$
- ▶ β_k does not depend on the dog $\Rightarrow$ no time by dog interaction

Split-plot ANOVA Table

Source	df	MS	F
Trt	$a - 1$	$MS_{trt} = \frac{SS_{trt}}{a-1}$	$F = \frac{MS_{trt}}{MS_{error_1}}$
Error 1	$N - a$	$MS_{error_1} = \frac{SS_{error_1}}{N-a}$	
Time	$t - 1$	$MS_{time} = \frac{SS_{time}}{t-1}$	$F = \frac{MS_{time}}{MS_{error_2}}$
Trt $\times$ Time	$(a - 1)(t - 1)$	$MS_{trt \times time} = \frac{SS_{trt \times time}}{(a-1)(t-1)}$	$F = \frac{MS_{trt \times time}}{MS_{error_2}}$
Error 2	$(N - a)(t - 1)$	$MS_{error_2} = \frac{SS_{error_2}}{(N-a)(t-1)}$	
Total	$Nt - 1$		

Dog Experiment Split-plot Analysis

```
> library(lmerTest)
> fit <- lmer(Response ~ Treatment * Time + (1 | Dog_id), data = dat)
> anova(fit)
```

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)	
Treatment	3.3396	1.11319	3	32	6.0038	0.002297	**
Time	6.2043	2.06811	3	96	11.1540	2.404e-06	***
Treatment:Time	3.4397	0.38219	9	96	2.0613	0.040573	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

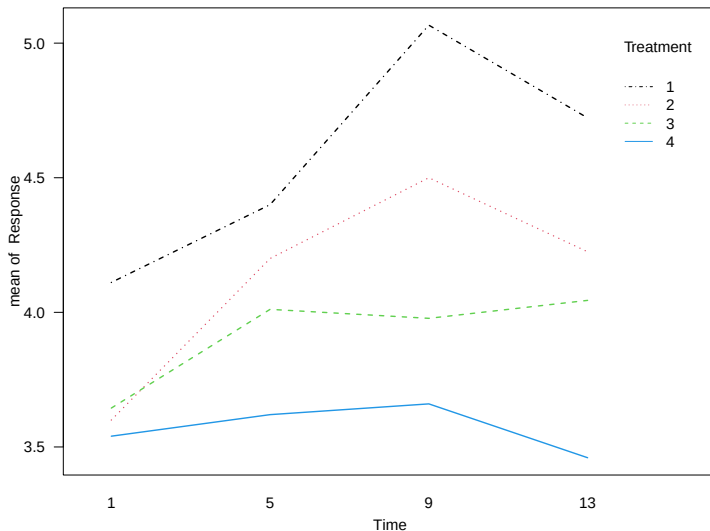
Hypothesis Tests:

We start with the interaction between treatment and time:

$$H_0 : (\alpha\beta)_{ik} = 0 \quad \forall i = 1, \dots, a, k = 1, 2, \dots, t$$

Result: We conclude the effect of treatment depends on time at $\alpha = 0.05$ level

Interaction Plot



Rejecting $H_0 : (\alpha\beta)_{ik} = 0$ means we reject the assumption of “parallelism”

Some Criticisms about the Split-ANOVA Approach

- ▶ The Split-plot ANOVA Approach assumes a constant correlation between any two observations from the same dog, that is, $\text{Cor}(Y_{ijk}, Y_{ijk'}) = \frac{\sigma_\delta^2}{\sigma_\delta^2 + \sigma_\varepsilon^2}$, this is the so-called **compound symmetry** correlation structure
- ▶ This assumption is unlikely to be valid with repeated measurements over time as the correlation for two nearby time points is likely to be higher than the correlation for two far apart time points
- ▶ Next, we are going to take a multivariate approach (MANOVA) as an attempt to address this issue

Compound Symmetry vs. AR(1)

Compound Symmetry

$$\mathbf{R}_{\text{CS}} = \begin{bmatrix} 1 & \rho & \rho & \rho \\ \rho & 1 & \rho & \rho \\ \rho & \rho & 1 & \rho \\ \rho & \rho & \rho & 1 \end{bmatrix}$$

All pairs have the same correlation.

AR(1)

$$\mathbf{R}_{\text{AR}(1)} = \begin{bmatrix} 1 & \rho & \rho^2 & \rho^3 \\ \rho & 1 & \rho & \rho^2 \\ \rho^2 & \rho & 1 & \rho \\ \rho^3 & \rho^2 & \rho & 1 \end{bmatrix}$$

Correlation decreases with temporal separation.

For measurements over time, AR(1) can represent the idea that **nearby observations are more strongly correlated**.

Approach 2: MANOVA

Here we consider the observations over time from the same dog, dog j receiving treatment i as a single vector of interest

$$\mathbf{Y}_{ij} = (Y_{ij1}, Y_{ij2}, \dots, Y_{ijt})^T,$$

and we will perform a one-way MANOVA

Assumptions:

- ▶ Dogs receiving treatment i have common mean vector $\boldsymbol{\mu}_i$
- ▶ All dogs have common covariance matrix $\boldsymbol{\Sigma}$
- ▶ Data from different dogs are independently sampled
- ▶ Data are multivariate normally distributed

Dog Experiment MANOVA Analysis

```
> dat <- read.table("dog1.txt")
> out <- manova(cbind(V3, V4, V5, V6) ~ as.factor(V1), data = dat)
> summary(out, test = "Wilks")
              Df    Wilks approx F num Df den Df  Pr(>F)
as.factor(V1)  3 0.48452    2.022    12 77.018 0.03316 *
Residuals      32
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Results: There are significant differences between at least one pair of treatments in at least one measurement of time

Criticism: MANOVA makes no assumptions regarding the temporal correlation structure, and hence, may be overparameterized leading to poor parameter estimates

Approach 3: Mixed Model Analysis

Main idea: Split-plot makes a too restrictive assumption while MANOVA makes no assumptions regarding the temporal correlation structure. The mixed model approach allows us to model the temporal correlation involving a limited number of parameters.

Model: $Y_{ijk} = \mu + \alpha_i + \delta_{j(i)} + \beta_k + (\alpha\beta)_{ik} + \varepsilon_{ijk}$.

Assumptions:

- ▶ $\varepsilon_{j(ik)} \stackrel{i.i.d.}{\sim} N(0, \sigma_\varepsilon^2)$
- ▶ $\delta_{j(i)} \stackrel{i.i.d.}{\sim} N(0, \sigma_\delta^2)$
- ▶ The correlation between the errors for the same dog depends only on the difference in observation time points: $|k - k'|$, e.g., $\text{Cor}(Y_{ijk}, Y_{ijk'}) = \rho^{|k-k'|}$ (Autoregressive with order 1)

Dog Experiment Mixed Model Analysis

```
> library(nlme)
> fit1 = gls(Response ~ Treatment * Time,
+           correlation = corCompSymm(form = ~ 1 | Dog_id), data = dat2)
> fit2 = gls(Response ~ Treatment * Time,
+           correlation = corAR1(form = ~ 1 | Dog_id), data = dat2)
> anova(fit1, fit2)
```

	Model	df	AIC	BIC	logLik
fit1	1	18	275.8063	327.1429	-119.9032
fit2	2	18	277.5811	328.9177	-120.7906

Results:

- ▶ Based on both AIC/BIC, having an AR(1) does not necessarily improve the model fit (in this data)
- ▶ However, having the option of modeling repeated measurement error structure can be useful in general as it provides additional modeling choices

Linking Repeated Measures to RRR

- ▶ Repeated measures can be seen as a **multivariate response vector**, just like in multivariate regression.
- ▶ Traditional **Split-plot ANOVA** assumes **compound symmetry** (constant correlation) $\Rightarrow$ often too restrictive.
- ▶ **MANOVA** removes correlation assumptions but can be **overparameterized**, especially with many time points.
- ▶ **Mixed models** allow more flexible structures (e.g., $AR(1)$) with fewer parameters.
- ▶ **Connection to RRR**: Just as RRR reduces dimensionality by finding a low-rank trajectory for correlated responses, repeated-measures methods aim to balance **parsimony vs. flexibility** in modeling time-dependent correlations.

Repeated Measures Analysis Summary

We learned about three approaches to analyze repeated measurements:

- ▶ Split-plot ANOVA
- ▶ MANOVA
- ▶ Mixed Effects Model

In the next lecture, we will learn about [Inference for Covariance Matrix](#)